Unit 19: Monopoly

I. Monopoly Profit Maximization

II. Market Power

Objectives of the Study:

After studying this unit, the student should be able to

-understand the Monopoly market structure

Monopoly

A monopoly is a market structure in which a single firm is the only supplier of a product that has no close substitutes. This means consumers do not have alternative choices, so the monopolist has significant control over the market. Monopolies have existed since ancient times. For example, the Greek philosopher Thales once controlled olive presses, and ancient Egyptian rulers-controlled food supply. In modern times, governments may also create monopolies by granting patents, which give firms exclusive rights to produce and sell a product for a certain period. Companies like Apple initially had near-monopoly power with products such as the iPod and iPad.

Unlike firms in perfect competition, a monopoly is a price maker, not a price taker. This means it has the power to influence or set the price of its product. Since the monopolist is the only seller, its output is equal to the total market supply, and it faces the market demand curve, which is downward sloping. This downward slope indicates that if the monopolist increases the price, the quantity demanded will decrease, but not fall to zero.

Because of this, a monopoly maximizes its profit by producing at the level where marginal revenue equals marginal cost (MR = MC) and then charging the highest price consumers are willing to pay for that quantity, based on the demand curve. As a result, the monopoly sets a price higher than marginal cost, unlike in perfect competition where price equals marginal cost.

This leads to important consequences:

- Consumers pay higher prices

- Quantity sold is lower than in a competitive market
- There is a loss of economic efficiency (deadweight loss)

Thus, monopoly results in reduced consumer welfare compared to perfect competition, as goods are underproduced and overpriced.

I. Monopoly Profit Maximization

All firms, including competitive firms and monopolies, maximize their profits by setting marginal revenue equal to marginal cost.

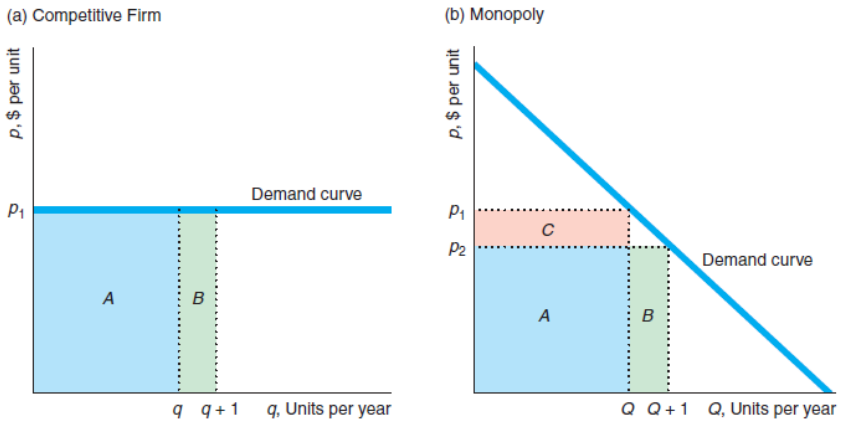
Marginal Revenue

A firm's marginal revenue curve depends on its demand curve. A monopoly's marginal revenue curve lies below its demand curve at any positive quantity because its demand curve is downward sloping.

Marginal Revenue and Price

A firm's demand curve shows the price, p, it receives for selling a given quantity, q. The price is the average revenue the firm receives, so a firm's revenue is R = pq. A firm's marginal revenue, MR, is the change in its revenue from selling one more unit. A firm that earns ΔR more revenue when it sells Δq extra units of output has a marginal revenue of MR = ΔR/Δq.

Figure 19.1: Demand Curve



If the firm sells exactly one more unit, Δq = 1, its marginal revenue is MR = ΔR. Although the marginal revenue curve is horizontal for a competitive firm, it is downward sloping for a monopoly. The competitive firm in panel a of figure 19.1 faces a horizontal demand curve at the market price, p_1. Because its demand curve is horizontal, the competitive firm can sell

another unit of output without dropping its price. As a result, the marginal revenue it receives from selling the last unit of output is the market price.

Initially, the competitive firm sells q units of output at the market price of p_1, so its revenue, R_1, is area A, which is a rectangle that is $p_1 \times q$. If the firm sells one more unit, its revenue is $R_2 = A + B$, where area B is $p_1 \times 1 = p_1$. The competitive firm's marginal revenue equals the market price:

$$\Delta R = R_2 - R_1 = (A + B) - A = B = p_1.$$

A monopoly faces a downward-sloping market demand curve, as in panel b of figure 19.1. (We've called the number of units of output a firm sells q and the output of all the firms in a market, or market output, Q. Because a monopoly is the only firm in the market, q and Q do not differ, so we use Q to describe both the firm's and the market's output.) The monopoly, which is initially selling Q units at p_1, can sell one extra unit only if the price falls to p_2.

The monopoly's initial revenue, $p_1 \times Q$, is $R_1 = A + C$. When it sells the extra unit, its revenue, $p_2 \times (Q + 1)$, is $R_2 = A + B$. Thus, its marginal revenue is

$$\Delta R = R_2 - R_1 = (A + B) - (A + C) = B - C.$$

The monopoly sells the extra unit of output at the new price, p_2, so its extra revenue is

$$B = p_2 \times 1 = p_2.$$

The monopoly loses the difference between the new price and the original price, $\Delta p = (p_2 - p_1)$, on the Q units it originally sold:

$$C = \Delta p \times Q.$$

Thus, the monopoly's marginal revenue, $B - C = p_2 - C$, is less than the price it charges by an amount equal to area C. The competitive firm in panel a does not lose an area C from selling an extra unit because its demand curve is horizontal. It is the downward slope of the monopoly's demand curve that causes its marginal revenue to be less than its price.

Marginal Revenue Curve

Thus, the monopoly's marginal revenue curve lies below the demand curve at every positive quantity. In general, the relationship between the marginal revenue and demand curves depends on the shape of the demand curve. For all linear demand curves, the relationship between the marginal revenue and demand curve is the same. The marginal revenue curve is a straight line that starts at the same point on the vertical (price) axis as the demand curve but has twice the slope of the demand curve, so the marginal revenue curve hits the horizontal (quantity) axis at half the quantity as the demand curve.

Deriving the Marginal Revenue Curve

To derive the monopoly's marginal revenue curve, we write an equation summarizing the relationship between price and marginal revenue that panel b of Figure 19.1 illustrates. For a monopoly to increase its output by ΔQ, the monopoly lowers its price per unit by Δp/ΔQ, which is the slope of the demand curve. By lowering its price, the monopoly loses (Δp/ΔQ) ×Q on the units it originally sold at the higher price (area C), but it earns an additional p on the extra output it now sells (area B). Thus, the monopoly's marginal revenue is

$$\mathrm{MR} = \mathrm{p} + \frac{\Delta \mathrm{p}}{\Delta \mathrm{Q}} \mathrm{Q}$$

Because the slope of the monopoly's inverse demand curve, Δp/ΔQ, is negative, the last term in the equation (Δp/ΔQ) Q, is negative. This equation confirms that the price is greater than the marginal revenue, which equals p plus a negative term.

Marginal Revenue and Price Elasticity of Demand

The marginal revenue at any given quantity depends on the demand curve's height (the price) and shape. The shape of the demand curve at a particular quantity is described by the price elasticity of demand, $\varepsilon = (\Delta Q/Q)/(\Delta p/p) < 0$, which is the percentage by which quantity demanded falls as the price increases by 1%.

At a given quantity, the marginal revenue equals the price times a term involving the elasticity of demand:

$$MR = p\left(1 + \frac{1}{\varepsilon}\right)$$

According to this equation, marginal revenue is closer to price as demand becomes more elastic. Where the demand curve hits the price axis (Q = 0), the demand curve is perfectly elastic, so the marginal revenue equals price: MR = p. Where the demand elasticity is unitary, $\varepsilon = -1$, marginal revenue is zero: MR = p[1 + 1/(-1)] = 0.

Marginal revenue is negative where the demand curve is inelastic, $-1 < \varepsilon \leq 0$.

Choosing Price or Quantity

Any firm maximizes its profit by operating where its marginal revenue equals its marginal cost. Unlike a competitive firm, a monopoly can adjust its price, so it has a choice of setting its price or its quantity to maximize its profit. (A competitive firm sets its quantity to maximize profit because it cannot affect market price.)

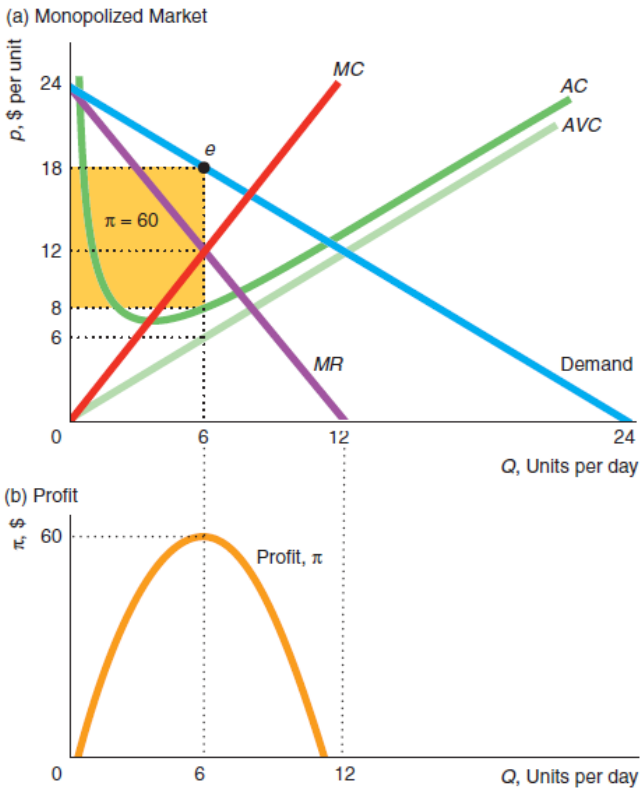
The monopoly is constrained by the market demand curve. Because the demand curve slopes down, the monopoly faces a trade-off between a higher price and a lower quantity or a lower price and a higher quantity. The monopoly chooses the point on the demand curve that

maximizes its profit. Unfortunately for the monopoly, it cannot set both its quantity and its price. If it could do so, the monopoly would choose an extremely high price and an extremely high output level—above its demand curve—and would become exceedingly wealthy. If the monopoly sets its price, the demand curve determines how much output it sells. If the monopoly picks an output level, the demand curve determines the price. Because the monopoly wants to operate at the price and output at which its profit is maximized, it chooses the same profit-maximizing solution whether it sets the price or output. In the rest of this chapter, we assume that the monopoly sets quantity

Graphical Approach

All firms, including monopolies, use a two-step analysis to determine the output level that maximizes their profit. First, the firm determines the output, Q*, at which it makes the highest possible profit—the output at which its marginal revenue equals its marginal cost. Second, the firm decides whether to produce Q* or shut down.

Figure 19.2: Monopoly Profit Maximisation



In the figure 19.2 Panel b shows the corresponding profit curve. The profit curve reaches its maximum at 6 units of output, where marginal profit—the slope of the profit curve—is zero.

Because marginal profit =marginal revenue - marginal cost, marginal profit is zero where marginal revenue equals marginal cost. In panel a, marginal revenue equals marginal cost at 6 units. The price on the demand curve at that quantity is 18. Thus, the monopoly maximizes its profit at point *e*, where it sells 6 units per day at a price of 18 per unit.

Why does the monopoly maximize its profit by producing where its marginal revenue equals its marginal cost? At smaller quantities, the monopoly's marginal revenue is greater than its marginal cost, so its marginal profit is positive—the profit curve is upward sloping. By increasing its output, the monopoly raises its profit. Similarly, at quantities greater than 6 units, the monopoly's marginal cost is greater than its marginal revenue, so its marginal profit is negative, and the monopoly can increase its profit by reducing its output. The marginal revenue curve is positive where the elasticity of demand is elastic, is zero at the quantity where the demand curve has a unitary elasticity, and is negative at larger quantities where the demand curve is inelastic. Because the marginal cost curve is never negative, the marginal revenue curve can only intersect the marginal cost curve where the marginal revenue curve is positive, in the range in which the demand curve is elastic. That is, a monopoly's profit is maximized in the elastic portion of the demand curve.

Shutdown Decision

A monopoly shuts down to avoid making a loss in the short run if its price is below its average variable cost at its profit-maximizing (or loss-minimizing) quantity. In the long run, the monopoly shuts down if the price is less than its average cost. In the short-run example in Figure 19.2, the average variable cost, AVC = 6, is less than the price, p = 18, at the profit-maximizing output, Q = 6, so the firm chooses to produce. Price is also above average cost at Q = 6, so the monopoly makes a positive profit. At the profit-maximizing quantity of 6 units, the price is P (6) = 18 and the average cost is AC (6) = 8. As a result, the profit, $\pi = 60$, is the golden rectangle with a height equal to the average profit per unit, p (6) – AC (6) = 18 - 8 = 10, and a width of 6 units.

Mathematical Approach

We can also solve for the profit-maximizing quantity mathematically. We already know the demand and marginal revenue functions for this monopoly. We need to determine its marginal cost curve. The monopoly's cost is a function of its output, C(Q). In Figure 19.2, we assume that the monopoly faces a short-run cost function of

$$C(Q) = Q^2 + 12$$

where Q^2 is the monopoly's variable cost as a function of output and 12 is its fixed cost. Given this cost function, the monopoly's marginal cost function is

$$MC = 2Q$$

This marginal cost curve is a straight line through the origin with a slope of 2 in panel a. The average variable cost is $AVC = Q^2/Q = Q$, so it is a straight line through the origin with a slope of 1. The average cost is $AC = C/Q = (Q^2 + 12)/Q = Q + 12/Q$, which is U-shaped.

We determine the profit-maximizing output by equating the marginal revenue and marginal cost functions:

$$MR = 24 - 2Q = 2Q = MC.$$

Solving for Q, we find that $Q = 6$. Substituting $Q = 6$ into the inverse demand function, we learn that the profit-maximizing price is

$$p = 24 - Q = 24 - 6 = 18.$$

At that quantity, the average variable cost is $AVC = 6$, which is less than the price, so the firm does not shut down. The average cost is $AC = 6 + 12/6 = 8$, which is less than the price, so the firm makes a profit.

II. Market Power

A monopoly has market power: the ability of a firm to charge a price above marginal cost and earn a positive profit. We now examine the factors that determine how much above its marginal cost a monopoly sets its price.

Market Power and the Shape of the Demand Curve

The degree to which the monopoly raises its price above its marginal cost depends on the shape of the demand curve at the profit-maximizing quantity. If the monopoly faces a highly elastic—nearly flat—demand curve at the profit-maximizing quantity, it would lose substantial sales if it raised its price by even a small amount. Conversely, if the demand curve is not very elastic (relatively steep) at that quantity, the monopoly would lose fewer sales from raising its price by the same amount.

We can derive the relationship between market power and the elasticity of demand at the profit-maximizing quantity using the expression for marginal revenue in equation

$$MR = p\left(1 + \frac{1}{\varepsilon}\right)$$

and the firm's profit-maximizing condition that marginal revenue equals marginal cost:

$$MR = p\left(1 + \frac{1}{\varepsilon}\right) = MC$$

By rearranging terms, we can rewrite this equation as

$$\frac{P}{MC} = \frac{1}{1 + \left({}^{1}/_{\varepsilon}\right)}$$

This equation says that the ratio of the price to marginal cost depends only on the elasticity of demand at the profit-maximizing quantity.

In our linear demand example in panel a of Figure 19.2, the elasticity of demand is ε = -3 at the monopoly optimum where Q = 6. As a result, the ratio of price to marginal cost is

p/MC = 1/ [1 + 1/ (-3)] = 1.5, or p = 1.5MC.

The profit maximizing price, 18, in panel a is 1.5 times the marginal cost of 12. If this monopoly were to raise its price, it would lose all its sales, so it maximizes its profit by setting its price equal to its marginal cost. The more elastic the demand curve at the optimum, the less a monopoly can raise its price without losing sales. All else the same, the closer substitutes for the monopoly's good, the more elastic is the demand curve at the optimum. For example, the publisher Pearson has the monopoly right to produce and sell microeconomics textbook. However, many other publishers have the rights to produce and sell similar microeconomics texts. The demand curve that Pearson faces is much more elastic than it would be if no substitutes were available.

Sources of Market Power

When will a monopoly face a relatively elastic demand curve and hence have little market power? Ultimately, the elasticity of demand of the market demand curve depends on consumers' tastes and options. The more consumers want a good—the more willing they are to pay "virtually anything" for it—the less elastic is the demand curve.

All else the same, the demand curve a firm (not necessarily a monopoly) faces becomes more elastic as (1) better substitutes for the firm's product are introduced, (2) more firms enter the market selling the same product, or (3) firms that provide the same service locate closer to this firm. The demand curves for Xerox, the U.S. Postal Service, and McDonald's have become more elastic in recent decades for these three reasons.

When Xerox started selling its plain-paper copier, no other firm sold a close substitute. Other companies' machines produced copies on special slimy paper that yellowed quickly. As other firms developed plain-paper copiers, the demand curve that Xerox faced became more elastic.

In the past, the U.S. Postal Service (USPS) had a monopoly in overnight delivery services. Now FedEx, United Parcel Service, and many other firms compete with the USPS in providing overnight deliveries. Because of these increases in competition, the USPS's share of business and personal correspondence fell from 77% in 1988 to 59% in 1996. Its total mail volume fell 40% from 2006 to 2010. Its share of the ground shipping market fell to 16% by 2012 (FedEx has about a third and UPS about half of the market). Compared to when it was a monopoly, the USPS's demand curves for first-class mail and package delivery have shifted downward and become more elastic.

As you drive down a highway, you may notice that McDonald's restaurants are located miles apart. The purpose of this spacing is to reduce the likelihood that two McDonald's outlets will compete for the same customer. Although McDonald's can prevent its own restaurants from competing with each other, it cannot prevent Wendy's or Burger King from locating near its restaurants. As other fast-food restaurants open near a McDonald's, that restaurant faces a more elastic demand. What happens as a profit-maximizing monopoly faces more elastic demand? It has to lower its price.

In conclusion, a monopoly maximizes its profit by producing at the level where marginal revenue equals marginal cost (MR = MC) and then charging a price based on the market demand curve, which is higher than marginal cost. This ability to set price arises from its market power, as it is the sole producer with no close substitutes. As a result, monopolies restrict output and charge higher prices compared to competitive markets. While this allows the firm to earn higher profits, it leads to reduced consumer welfare and inefficiency in resource allocation.

Multiple Choice Questions

1. In a monopoly market structure, what is the primary characteristic that distinguishes it from perfect competition?

   A. There are multiple firms supplying identical products.

   B. A single firm is the only supplier of a product with no close substitutes.

C. Prices are determined by market demand and supply.

D. Consumer choices are abundant.

Answer: B. A single firm is the only supplier of a product with no close substitutes.

2. What happens to consumer welfare in a monopoly compared to a competitive market?

A. Consumer welfare increases because of lower prices.

B. Consumer welfare is unaffected by monopoly pricing.

C. Consumer welfare decreases due to higher prices and lower output.

D. Consumer welfare remains the same as there are no substitutes.

Answer: C. Consumer welfare decreases due to higher prices and lower output.

3. What is the result of a monopolist setting a price above marginal cost?

A. Increased consumer surplus and higher demand.

B. A deadweight loss due to inefficiency in resource allocation.

C. Greater production efficiency compared to competitive markets.

D. An increase in the quantity sold.

Answer: B. A deadweight loss due to inefficiency in resource allocation.

4. What is the shape of a monopoly's marginal revenue curve compared to its demand curve?

A. The marginal revenue curve is above the demand curve.

B. The marginal revenue curve is the same as the demand curve.

C. The marginal revenue curve lies below the demand curve.

D. The marginal revenue curve is horizontal across all quantities.

Answer: C. The marginal revenue curve lies below the demand curve.

5. How does a monopoly maximize its profit?

   A. By producing at the level where marginal cost equals average revenue.

   B. By producing at the level where marginal revenue equals marginal cost.

   C. By selling as much as possible at the market price.

   D. By setting prices equal to marginal cost.

   Answer: B. By producing at the level where marginal revenue equals marginal cost.